From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (08 – 11) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 07)** contains 7 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

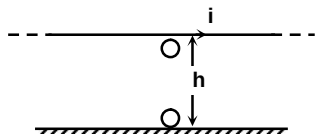
Physics

PART – I

SECTION – A
Single Correct Choice Type

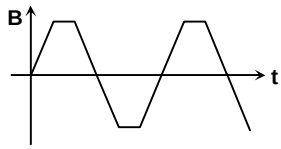
This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

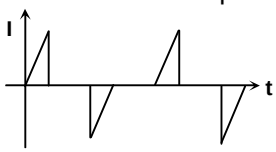
1. A small circular coil of radius 0.5 cm is held directly below a long straight wire carrying a current of 10.0 A. The plane of coil lies in the plane of paper. The coil is located 0.5 m above the top of a table. Now the coil is dropped from rest what is the magnitude of the average induced emf approximately in the coil from the time it is released to the moment it hits the top of the table. Assume that the magnetic field is nearly constant over the area of the coil and is equal to magnetic field at the centre of the coil. ($g = 10\ m/s^2$

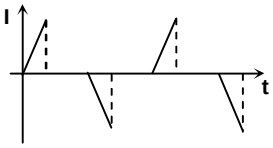


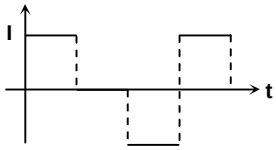
(A) 10^{-8} V (B) 10^{-6} V
(C) 10^{-5} V (D) 10^{-9} V

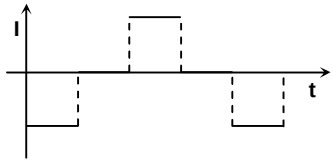
2. Magnetic field through the loop varies as shown in the figure. (Positive magnetic field implies magnetic field perpendicular to the plane of the loop and assume anticlockwise current as positive).



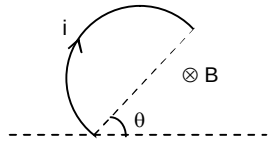
(A) 

(B) 

(C) 

(D) 

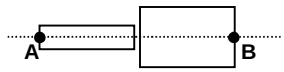
3. A semicircular ring of radius R carrying current i is placed in a magnetic field of intensity B so that plane of wire is perpendicular to magnetic field as shown. Net force acting on the ring is



(A) 2BiR (B) 2BiRsin θ
(C) 2BiRcos θ (D) BiR sin θ

Space for rough work

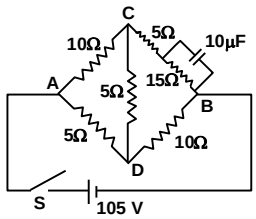
4. A cylindrical element of resistivity ρ, radius r and length ℓ is connected in series with another cylindrical element of conductivity σ, radius 2r and length ℓ, such that axis of both the cylinders is along the same line as shown. The net resistance across A and B is



(A) $\left(\rho+\frac{1}{4\sigma}\right)\frac{\ell}{\pi r^2}$ (B) $\left(\rho+\frac{4}{\sigma}\right)\frac{\ell}{\pi r^2}$

(C) $\left(\frac{1}{\rho}+4\sigma\right)\frac{\pi r^2}{\ell}$ (D) $\left(4\rho+\frac{1}{\sigma}\right)\frac{2\ell}{\pi r^2}$

5. In the circuit shown, switch S is closed at t = 0. The current in the branch BD at t = 0 for the given circuit in A is (Assume capacitor to be uncharged initially)



(A) 7 (B) 9

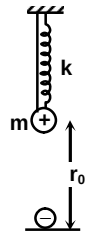
(C) 6 (D) 3

6. A galvanometer has sensitivity 35 kΩ/V and internal resistance 20 Ω. To convert it to an ammeter that reads 2A full scale a resistance of

(A) 2.9×10^{-4} Ω should be added in parallel. (B) 4 mΩ should be added in parallel.

(C) 2.9×10^{-4} Ω should be added in series. (D) 4 mΩ should be added in series.

7. A positively charged sphere of mass m = 5 kg is attached to one end of an ideal spring of spring constant k = 10^4 Nm^{-1} whose other end is fixed to a rigid support. The sphere is tied with a thread so that the spring is in its natural length. Another negatively charged sphere is fixed with floor, vertically below the positively charged sphere as shown. If initial separation between the spheres is 50 cm and magnitude of charge on each sphere is q = 100 μC. The maximum elongation of the spring when the thread is burnt is (g = 10 m/s^2)



(A) 10 cm (B) 15 cm

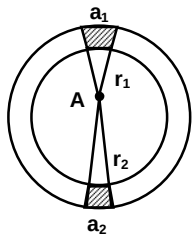
(C) 5 cm (D) 25 cm

Space for rough work

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

8. A wire having uniform positive charge density λ, is bent in the form of a ring of radius R. Point A as shown in the figure is in the plane of the ring but not on the centre. Two elements of ring of lengths a_1 and a_2 subtend very small same angle at point A. They are at distances r_1 and r_2 from the point A respectively ($r_2 > r_1$). Then



(A) The ratio of charges of elements a_1 and a_2 is $\frac{q_1}{q_2} = \frac{r_1}{r_2}$

(B) The element a_1 produces greater magnitude of electric field at A than a_2 ($E_1 > E_2$)

(C) $V_1 = V_2$

(D) At point A net electric field is towards a_2.

9. A charged particle is released in a gravity free space with some initial velocity in a magnetic field $\vec{B} = B_x\hat{i} + B_y\hat{j}$. If at any instant velocity of the particle is $\vec{v} = v_x\hat{i} + v_y\hat{j}$

(A) $v_x^2 + v_y^2$ is always a constant.

(B) If at any instant acceleration vector of the particle is $\vec{a}$, then $\vec{a}\cdot\vec{v} = 0$

(C) Tangential acceleration of the particle is always zero.

(D) Component of acceleration along x-axis is always zero.

10. Current flows through a straight cylindrical conductor of radius r. The current is distributed uniformly over its cross-section. The magnetic field at a distance x from the axis of the conductor has magnitude B.

(A) B = 0 at the axis.

(B) $B \propto x$ for $0 \le x \le r$.

(C) $B \propto \frac{1}{x}$ for x > r

(D) B is maximum for x = r.

11. Select the correct statement(s) from the following.
The path of a charged particle moving with an initial velocity u

(A) parallel to the uniform electrostatic field is a straight line.

(B) perpendicular to the uniform electrostatic field is a parabola.

(C) parallel to the uniform magnetic field is a helix with its axis parallel to the direction of B.

(D) perpendicular to the uniform magnetic field is a circle in a plane normal to the field.

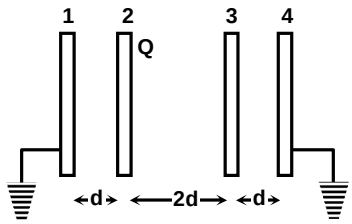
Space for rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 12 to 13

Four metallic plates are placed as shown in the figure. Plate 2 is given a charge Q whereas all other plates are uncharged. Plates 1 and 4 are earthed. The area of each plate is same.



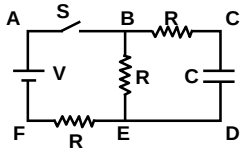
12. The charge appearing on the right side of plate 3 is
(A) zero (B) +Q/4
(C) –3Q/4 (D) Q/2

13. The charge appearing on right side of plate 4 is
(A) zero (B) –Q/4
(C) –3Q/4 (D) –Q/2

Space for rough work

Paragraph for Question Nos. 14 to 16

In the circuit shown, the battery is ideal with emf V. The capacitor is initially uncharged. The switch S is closed at t = 0



14. The charge on the capacitor at time t is

(A) $\frac{VC}{2}(1-e^{-2t/3RC})$ (B) $\frac{VC}{2}(1-e^{-t/3RC})$

(C) $\frac{VC}{3}(1-e^{-2t/3RC})$ (D) $\frac{VC}{3}(1-e^{-t/3RC})$

15. The current in ED at time t is

(A) $\frac{V}{3R}e^{-2t/3RC}$ (B) $\frac{V}{2R}e^{-2t/3RC}$

(C) $\frac{V}{3R}e^{-t/3RC}$ (D) $\frac{V}{2R}e^{-t/3RC}$

16. The current in EB at time t is

(A) $\frac{V}{6R}(3e^{-2t/3RC}-1)$ (B) $\frac{V}{2R}$

(C) $\frac{Ve^{-2t/3RC}}{R}$ (D) $\frac{V}{6R}(e^{-2t/3RC}-3)$

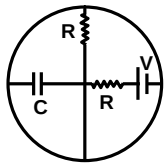
SECTION –C
Integer Answer Type

This section contains **7 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. Find the minimum number of capacitors of 2μF capacitance each, required to obtain a capacitance of 5μF.

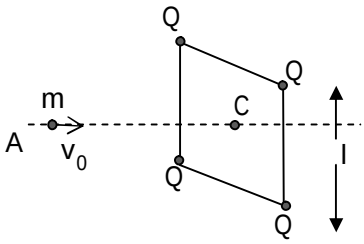
Space for rough work

2. Charge on capacitor in steady state is (in Volt)

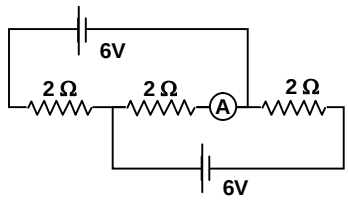


3. The resistance of a series combination of two resistances is S. When they are joined in parallel, the equivalent resistance is P. Find the minimum value of $\frac{S}{P}$.

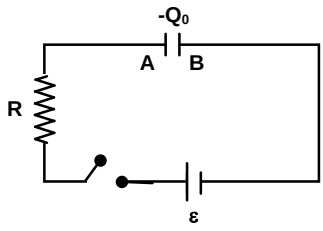
4. Four identical point charges are fixed at the four corners of a square of a side length l. Another charged particle of mass m and charge +q is projected towards centre of square from a large distance along the line perpendicular to plane of square. The minimum value of initial velocity v_0 required to cross the square is?
(m = 1 gm, l = $4\sqrt{2}$ m, Q = 1 μc, q = 0.5 μc)



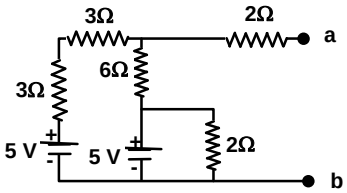
5. What is the reading of ideal ammeter (in A) in the circuit shown in the adjacent figure?



6. The figure shows a RC circuit with a parallel plate capacitor. Before switching on the circuit, plate A of the capacitor was given a charge $-Q_0$ while plate B had no net charge.
Now, at t = 0, the circuit is switched on. How much time (in seconds) will elapse before the net charge on plate A becomes zero. Take C = 1 μF, Q_0 = 1 mC, ε = 1000 V and R = $\frac{2\times10^6}{\ell n(3)}\Omega$



7. Determine the resistance (in Ω) to be connected across a and b in given circuit to dissipate maximum power through it.



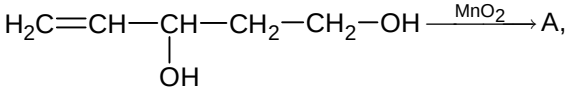
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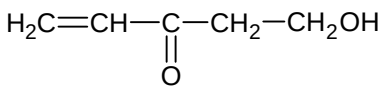
Chemistry

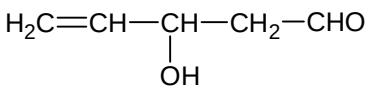
PART – II

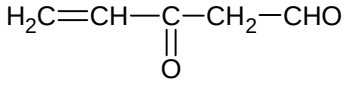
SECTION – A
Single Correct Choice Type

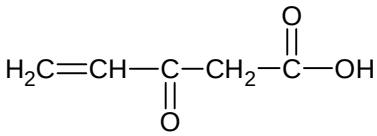
This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. 
A is

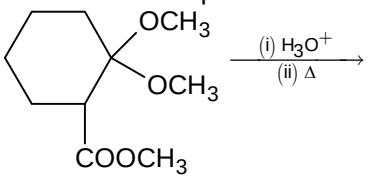
(A) 

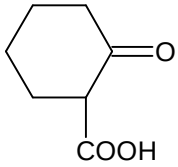
(B) 

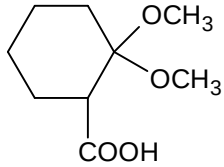
(C) 

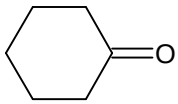
(D) 

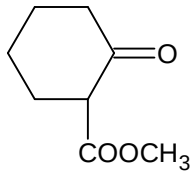
2. What is the final product of the following reaction?



(A) 

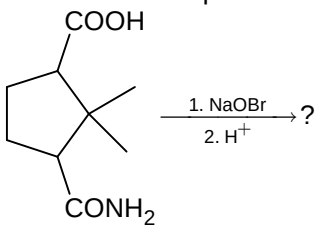
(B) 

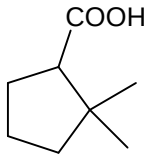
(C) 

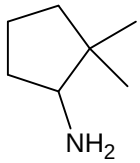
(D) 

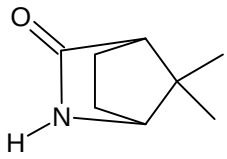
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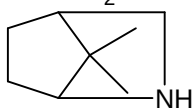
3. What is the final product of the following reaction?

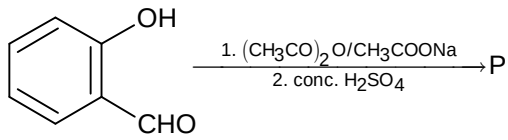


(A) 

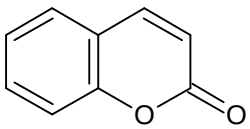
(B) 

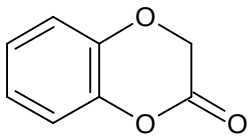
(C) 

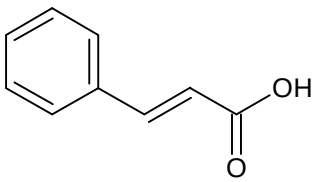
(D) 

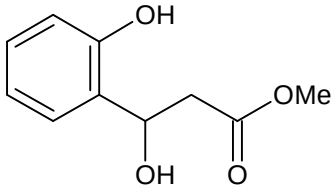
4. 

What is the product P?

(A) 

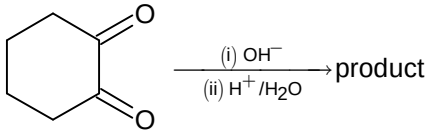
(B) 

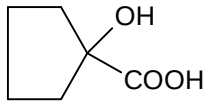
(C) 

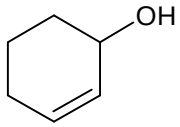
(D) 

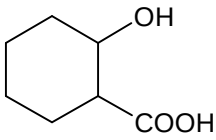
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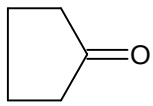
5. Predict the product in the following reaction:



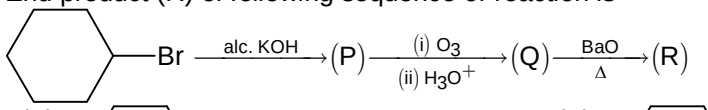
(A) 

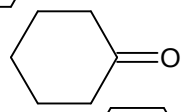
(B) 

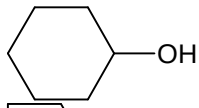
(C) 

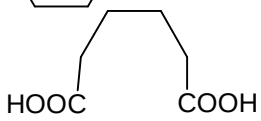
(D) 

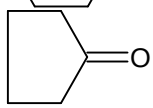
6. End product (R) of following sequence of reaction is



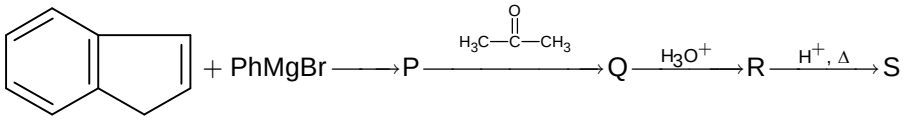
(A) 

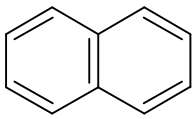
(B) 

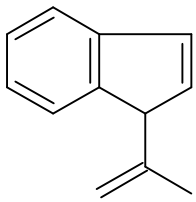
(C) 

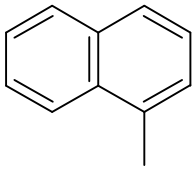
(D) 

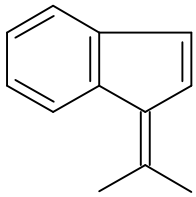
7. Identify S in the following reaction:



(A) 

(B) 

(C) 

(D) 

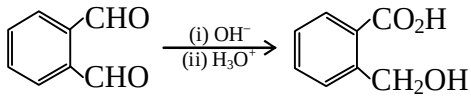
Space for rough work

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

8. Select the correct statement(s), among the following given statements
   (A) $CH_3CH_2CH_2I$ will react more readily than $(CH_3)_2CHI$ for SN^2 reactions
   (B) $CH_3CH_2CH_2Cl$ will react more readily than $CH_3CH_2CH_2Br$ for SN^2 reactions
   (C) $CH_3CH_2CH_2CH_2Br$ will react more readily than $(CH_3)_3C - CH_2Br$ for SN^2 reactions
   (D) $CH_3 - O - C_6H_5 - Br$ will react more readily than $NO_2 - C_6H_5 - CH_2Br$ for SN^2 reaction

9. Consider following reaction



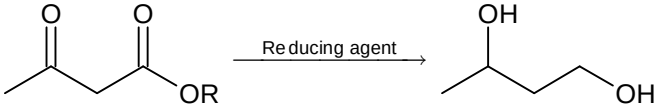
   This:
   (A) is a crossed Cannizzaro reaction.
   (B) is intramolecular Cannizzaro reaction.
   (C) causes substitution of H^- by OH^- in one–CHO group and is thus oxidised while second –CHO undergoes an addition of H^- and is thus reduced to alcohol.
   (D) resembles aldol type condensation reaction.

10. Lucas test is used to make distinction between 1°, 2° and 3° alcohols:

$$ROH + HCl(conc.) \xrightarrow{\text{anhy. } ZnCl_2} \underset{(\text{White turbidity})}{RCl\downarrow} + H_2O$$

   This show that
   (A) ROH behaves as a base
   (B) Greater the value of pK_a(alcohol), greater the reactivity with conc. HCl & thus sooner the formation of white turbidity
   (C) Alcohol which reacts fastest with Na metal, will give turbidity at fastest rate
   (D) Alcohol which gives red colour during victor mayor test will give turbidity at slower rate in Lucas test than those giving blue or white colour during victor mayor test

11. Reducing agent(s) which can be used to carry out following conversion is/are



   (A) $NaBH_4$ (B) $LiAlH_4$
   (C) H_2/Ni (D) B_2H_6/THF

Space for rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 12 to 13

Aldehydes without α - hydrogen undergoes Cannizzaro reaction in the presence of conc. Alkali. Two molecule undergoes self oxidation and reduction to yield a mixture of alcohol and salt of carboxylic acid. Two different aldehydes without α hydrogen undergoes crossed Cannizzaro reaction. The rate of reaction is decided by the ease with which H^- is transferred.

12. $HCHO + PhCHO \xrightarrow{50\%\ NaOH} P + Q$

P and Q are

(A) HCOOH and PhCOONa
(B) HCOONa and PhCOOH
(C) CH_3OH and PhCOONa
(D) HCOONa and $PhCH_2OH$

13. $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-H \xrightarrow{conc.\ NaOH} Product(s)$

Choose the correct statements among the following statement:

(A) product are $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-COONa$ and $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-CH_2OH$

(B) Product is $(H_3C)_3C-\overset{OH}{\underset{|}{C}}-\overset{O}{\overset{\|}{C}}-ONa$ and reaction in disproportionation reaction

(C) Product is $(H_3C)_3C-\overset{OH}{\underset{|}{C}}-\overset{O}{\overset{\|}{C}}-ONa$ but is not disproportionation reaction

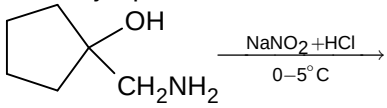
(D) Product are $(H_3C)_3C-\overset{OH}{\underset{|}{C}}-CH$ and $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-CH_2OH$

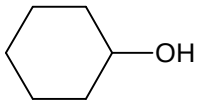
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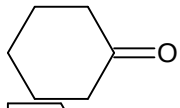
Paragraph for Question Nos. 14 to 16

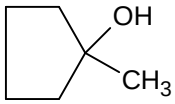
Aliphatic primary amines react with nitrous acid ($NaNO_2$ + HCl) form alcohol as the major product. In addition to alcohol, alkenes and alkyl halide are also formed as minor products. Nitrous acid also react with secondary and tertiary amine, nature of reaction depends whether amine is aliphatic or aromatic.

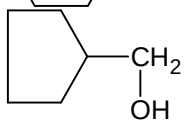
14. The major product obtained in the following reaction will be



(A) 

(B) 

(C) 

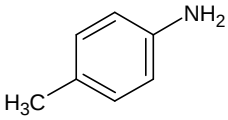
(D) 

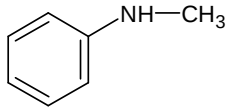
15. Which of the following product(s) will be obtained when n-propyl amine is treated with $NaNO_2$ and HCl?

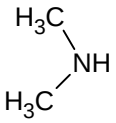
(A) $H_3C-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$ (B) $H_3C-CH=CH_2$

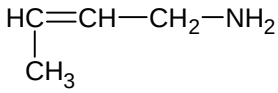
(C) $H_3C-\underset{\displaystyle Cl}{\underset{|}{CH}}-CH_3$ (D) All of these

16. Which of the following amine evolve N_2 gas with $NaNO_2$/HCl temperature of reaction below 5°C?

(A) 

(B) 

(C) 

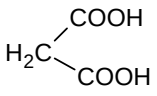
(D) 

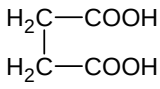
SECTION –C

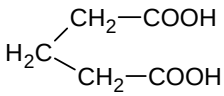
Integer Answer Type

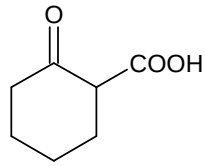
This section contains **7 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

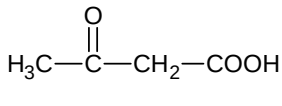
1. How many of the following acids easily undergo decarboxylation on heating?





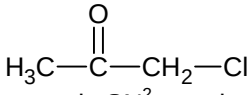




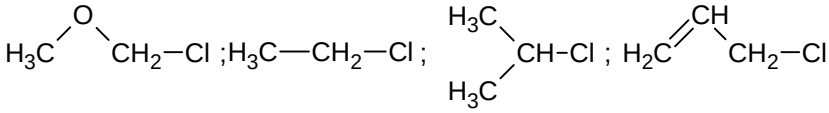


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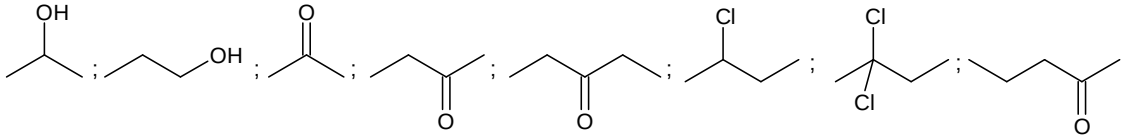
2. How many of the following compounds are less reactive than



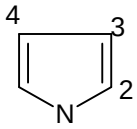
towards SN^2 reaction:



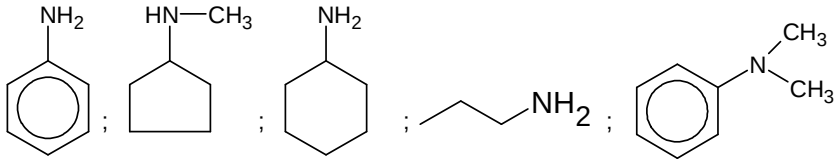
3. How many of the following compounds give positive haloform test?



4. Which of the following mentioned position in the given compound is more reactive towards electrophilic substitutions?

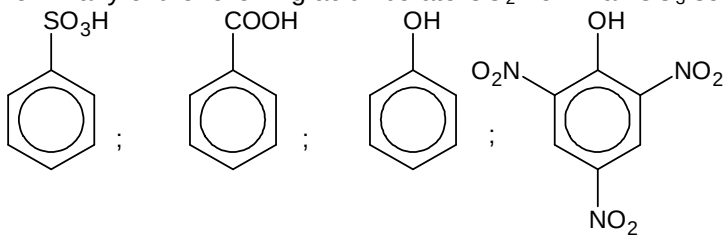


5. How many of the following give positive carbyl amine test?



6. How many moles of phenyl hydrazine react with glucose to form glucosazone?

7. How many of the following acid liberate CO_2 from $NaHCO_3$ solution?



Space for rough work

Mathematics

PART – III

SECTION – A
Single Correct Choice Type

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. If $f(x) = (-1)^{\left[\frac{2x}{\pi}\right]}$, $g(x) = |\sin x| - |\cos x|$ and $\phi(x) = f(x)\, g(x)$ (where [.] denotes the greatest integer function) then the respective fundamental periods of f(x), g(x) and $\phi(x)$ are

(A) π, π, π (B) $\pi, 2\pi, \pi$

(C) $\pi, \pi, \frac{\pi}{2}$ (D) $\pi, \frac{\pi}{2}, \pi$

2. $\int_0^{\pi/2} \sin x \, \log(\sin x)\, dx$ is equal to

(A) $\log_e(e/2)$ (B) $\log 2 - e$

(C) $\log_e((2/e)$ (D) $\log e - 2$

3. The value of $\int f''(x)F(x)\,dx - \int f(x)F''(x)\,dx$ is equal to

(A) $f(x)\,F(x) - f''(x)\,F''(x) + c$ (B) $f'(x)\,F(x) - f(x)\,F'(x) + c$

(C) $f(x)\,F'(x) - F(x)\,f'(x) + c$ (D) $f'(x)\,F''(x) - F'(x)\,f''(x) + c$

Space for rough work

4. If $f(x) = ax^2 + bx + c$ is such that $|f(x)| \le 1$, $\forall\, x \in [-1, 1]$ and $g(x) = |f(x)|$, $x \in [1, \infty]$, then $\int_1^2 g(x)\,dx$ is

(A) less than or equal to $\frac{11}{3}$ (B) greater than or equal to $\frac{1}{3}$

(C) less than or equal to $-\frac{5}{3}$ (D) lies between $-\frac{1}{3}$ to $\frac{5}{3}$

5. Let $f\left(-\frac{\pi}{2}\right) = -1$, $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 1$ be a continuous and twice differentiable function, then

(A) $|f''(x)| \le 1\ \forall\, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $|f''(x)| = 1$ for atleast one $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(C) $|f''(x)| \ge 1\ \forall\, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (D) $|f''(x)| < 1$ for atleast one $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

6. $\int \frac{x^3 - 1}{x^4\sqrt{6x^6 - 4x^3 + 2}}\,dx$ equals $(x > 0)$

(A) $\frac{\sqrt{6x^6 - 4x^3 + 2}}{12x^3} + c$ (B) $\frac{\sqrt{6x^6 - 4x^3 + 2}}{12x^2} + c$

(C) $\frac{\sqrt{6x^6 - 4x^3 + 2}}{6x^3} + c$ (D) $\frac{\sqrt{6x^6 - 4x^3 + 2}}{6x^2} + c$

7. The value of $\int_0^{\pi/3} \left[\sqrt{3}\tan x\right]dx$ (where [.] denotes the greatest integer function) is

(A) $\cot^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (B) $\frac{5\pi}{6} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(C) $\frac{\pi}{6} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) none of these

Space for rough work

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

8. The value of $\lim_{x\to -4}\frac{\tan \pi x}{x+4}+\lim_{x\to\infty}\left(1+\frac{1}{x^2}\right)^x$ is
(A) greater than 3 (B) greater than 4
(C) greater than 5 (D) less than 6

9. If [x] represents the greatest integer less than or equal to x, then which of the following statement is true
(A) sin [x] = cos [x] has no solution
(B) sin [x] = tan [x] has infinitely many solutions
(C) sin [x] = cos [x] possess unique solution
(D) sin [x] = tan [x] for no value of x

10. Let f and g be functions from the interval [0, ∞) to the interval [0, ∞) f being an increasing function and g being a decreasing function. If f {g (0)} = 0, then
(A) f {g (x)} ≥ f {g (0)} (B) g {f (x)} ≤ g {f (0)}
(C) f {g (1)} = 0 (D) none of these

11. If f(g(x)) is a one-one function, f: R → R and g: R → R, then
(A) g(x) must be one-one (B) f(x) must be one-one
(C) f(x) may not be one-one (D) g(x) may not be one-one

Space for rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 12 to 13

Read the following write up carefully and answer the following questions:

If f: R $\rightarrow$ R and f(x) = g(x) + h(x) where g(x) is a polynomial and h(x) is a continuous and differentiable bounded function on both sides, then f(x) is onto if g(x) is of odd degree and f(x) is into if g(x) is of even degree. To check whether f(x) is one-one we need to differentiate f(x). If f′(x) changes sign in domain of f then f is many-one else one-one

12. f: R $\rightarrow$ R and $f(x) = a_1x + a_3x^3 + a_5x^5 + ... + a_{2n+1}x^{2n+1} - \cot^{-1}x$ where $0 < a_1 < a_3 < ... < a_{2n+1}$, then the function f(x) is
    (A) one-one into (B) many-one onto
    (C) one-one onto (D) many-one into

13. f: R $\rightarrow$ R and $f(x) = \frac{x(x^4+1)(x+1)+x^4+2}{x^2+x+1}$, then f(x) is
    (A) one-one into (B) many-one onto
    (C) one-one onto (D) many-one into

Space for rough work

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:
Let the curve f(x, y) = 0, where $x = 1 - 3t^2$, $y = t - 3t^3$. If the tangent at the point P(t) is drawn, such that it makes an angle θ to the positive direction of x-axis.

14. For the point P(1), if m denotes the slope of the tangent then the value of $m + \sqrt{1+m^2}$ is
(A) 0 (B) 1
(C) 2 (D) 3

15. If the tangent at P(–1) meet the curve again at $Q(t_1)$ then the coordinates of point Q be
(A) $\left(-3, -\frac{9}{2}\right)$ (B) $\left(-\frac{1}{3}, -\frac{2}{9}\right)$
(C) $\left(-\frac{1}{3}, \frac{2}{9}\right)$ (D) $\left(\frac{1}{3}, -\frac{2}{9}\right)$

16. If the tangent at P(1) meets the curve again at Q, then the angle between the tangent at P and Q is
(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Rough work

SECTION –C

Integer Answer Type

This section contains **7 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. $\frac{1}{8}\int 7\left\{\frac{x^2(x^4-4x-3)}{(x^3-1)^2}\cos x-\frac{(x^4+1)}{(x^3-1)}\sin x\right\}dx$ is equal to _____

2. $I=\int_0^{1.5}\left[x^2\right]dx$ where [.] is greatest integer function then the value of [I] is _____

3. If α is the absolute maximum value of the expression $\frac{3x^2+2x-1}{x^2+x+1}\ \forall\ x\in R$, then $[\alpha]$ is _____, (where [.] denotes the greatest integer function)

4. The value of $\int_0^1 x\log x\,dx = -\frac{1}{k}$, then k is _____

5. If $f(x)=3\sin\sqrt{\frac{\pi^2}{9}-x^2}+\left||x|-\frac{3\sqrt{3}}{2}\right|$, then the maximum value of $\frac{f(x)}{\sqrt{3}}$ is _____

6. If f(x) is a real valued function discontinuous only at x = 0; having local extrema at x = 0 and satisfying $(f(x))^2=4\ \forall$ x, then the number of function f(x) is _____

7. Let f: R $\to$ R and g: R $\to$ R be two one-one and onto functions such that they are the mirror images of each other about the line y = 2. If h(x) = f(x) + g(x) then h(0) equals to _____

Rough work